

Business Development Companies (BDCs), Direct Lending, and Performance Dynamics in U.S. Middle-Market Credit

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1 Literature Review

After the global financial crisis, the financing landscape of the U.S. middle market underwent a structural shift. On the one hand, tighter regulation and higher capital requirements led traditional banks to retrench in both risk-taking capacity and business scope, particularly in lending to small and middle-market firms. However, the demand for flexible, relationship-based financing from these firms did not diminish; instead, amid increasing competition and increasingly complex business models, their financing needs became more diverse and less standardized. The resulting gap between constrained bank supply and persistent credit demand created room for non-bank lenders to expand, with Business Development Companies (BDCs) emerging as one of the most visible public-facing vehicles in this transition [Davydiuk et al., 2024b] [Kuan, 2025][Corwell et al., 2025]. In this sense, the significance of BDCs lies not only in their participation in direct lending, but also in their role in embedding what was traditionally a relatively opaque segment of private credit into public capital markets. By operating as publicly traded closed-end investment companies, BDCs make the performance, risk, and valuation of middle-market direct lending observable and continuously priced. As a result, they serve as a public window into the expansion of direct lending and private credit in the U.S. middle market.

Compared with traditional banks, BDCs differ markedly in their funding structure, regulatory framework, and interface with capital markets. [Davydiuk et al., 2024c] describe BDCs as a special type of closed-end investment company, whose funding model closely overlaps with broader direct-lending and private-debt activities, yet remains distinct from the liability structure and regulatory regime governing banks. This hybrid intermediary position is central to understanding BDC performance: outcomes are not determined solely by credit underwriting and loan recoveries but are also shaped by public-market valuation, investor

sentiment, and pricing frictions. [Kuan, 2025] further reinforces this point by highlighting the divergence between NAV-based returns and market-price dynamics. BDC performance thus inherently encompasses two dimensions: asset-side outcomes reflected in net asset value (NAV), and the resistance of risk and future cash flows in public trading markets. Consequently, the pronounced performance dynamics observed in BDCs do not arise merely from fluctuations in underlying asset returns, but from their simultaneous embedding in the credit mechanism of private lending and the valuation mechanism of public capital markets. These two mechanisms may reinforce or offset each other under different macroeconomic conditions and sentiment regimes, giving rise to more complex and state-dependent patterns in BDC performance.

With respect to performance dynamics, a central finding in the existing literature is that BDC performance does not converge to a single, stable “industry-wide alpha,” but instead exhibits substantial heterogeneity and dispersion. [Suhonen, 2024] documents significant cross-sectional variation in BDC and direct-lending performance, along with a degree of persistence over time. This suggests that focusing on “average direct-lending returns” or “average BDC performance” may obscure economically meaningful differences. The more relevant question, therefore, is why different BDCs display sustained performance differentials and how these differences evolve over time. In explaining the sources of such heterogeneity, [Kuan, 2025] emphasizes the importance of portfolio composition, highlighting in particular the role of senior secured loan allocation, size heterogeneity, and market sentiment as potential drivers of variation in NAV-based total returns. This perspective aligns with broader industry-oriented analyses suggesting that BDC outcomes depend on asset mix, leverage and liability structure, as well as the ability to manage frictions arising from public-market trading [Kuan, 2025][Warburton, 2020]. From a performance-dynamics perspective, therefore, BDC outcomes should not be viewed as static asset returns, but as the result of an evolving process shaped jointly by portfolio choices, funding structure, and capital-market pricing.

To further deepen the understanding of this process, recent research in private credit has shifted attention from “what the assets look like” to “who the borrowers are, how contracts are structured, and how market discipline operates.” [Jang et al., 2025]’s study on lending technology shows that direct lenders serve a systematically different borrower segment from traditional banks and finance companies—particularly firms that are younger, more intangible-intensive, and more closely connected to private equity (PE) sponsors [Jang et al., 2025]. This implies that BDC portfolio choices involve not only asset categories, but also the borrower ecosystem and informational environment into which they extend credit, thereby shaping default risk, contractual governance, and cash-flow stability. At the same time, [Davydiuk et al., 2024c] provide evidence on middle-market direct lenders and BDCs, emphasizing the real economic effects of financing access. This further reinforces the view that BDCs are not merely substitutes for banks, but participants in reshaping capital access and financing constraints for a particular segment of firms [Davydiuk et al., 2024b]. Within this framework, performance dynamics are reflected not only in market-price

fluctuations, but also in evolving credit outcomes driven by changes in borrower composition, contract design, and portfolio structure over the business cycle.

In explaining how BDC risk and returns evolve with macroeconomic conditions, interest rate sensitivity constitutes a particularly important, but relatively fragmented—line of research. Park characterizes BDCs as hybrid entities positioned between financial institutions and high-dividend vehicles, and finds that their stock returns exhibit measurable sensitivity to changes in interest rates. Park also discusses balance sheet channels and net interest margin (NIM) effects, suggesting that interest rate fluctuations may influence BDC performance through both funding costs and income structures, while leaving the precise mechanisms open for further investigation [Park, 2013]. From a performance-dynamics perspective, the key implication is that interest rates should not be viewed merely as an exogenous macroeconomic backdrop, but rather as a first-order determinant that can jointly drive valuation and operating performance. When changes in interest rates simultaneously affect loan side yields, liability side financing costs, and public-market discount rates, BDC returns and risk exposures become inherently state-dependent. In different interest rate regimes, the strength of the linkage between asset-side performance and market-side pricing may therefore shift systematically.

As the direct lending market has expanded, the focus of the literature has shifted from the traditional manager–investor conflict toward incentive arrangements within the capital structure and the role of market discipline. Existing evidence suggests that, in the direct lending space, conflicts and risk reallocation can arise across different layers of the capital structure, while creditor oversight and financing constraints may impose meaningful discipline on risk-taking [Davydiuk et al., 2024a][Davydiuk et al., 2024c]. This implies that risk choice is not determined solely by the available investment opportunity set or asset class, but is endogenously shaped by incentive structures and institutional constraints. In the context of BDCs, this consideration is particularly salient. As publicly traded direct lending intermediaries, BDCs are subject to both internal incentive arrangements and external market discipline. Consequently, even when asset portfolios appear similar, differences in governance structures and financing discipline may lead BDCs to exhibit systematic variation in risk intensity, leverage usage, and cyclical adjustment behavior. Governance mechanisms, therefore, are not peripheral to asset allocation decisions, but rather constitute a central condition shaping how risk is selected, amplified, or constrained.

Overall, the existing literature has examined BDC performance and risk in U.S. middle-market direct lending from multiple dimensions, including portfolio composition, borrower characteristics, interest rate conditions, and governance mechanisms [Kuan, 2025][Suhonen, 2024][Jang et al., 2025][Davydiuk et al., 2024c]. These studies suggest that BDC returns reflect not only the operating outcomes of underlying credit assets, but also the influence of capital market pricing mechanisms. Similarly, risk exposure is shaped both by portfolio choices and by macroeconomic interest rate regimes and governance constraints. However, much of the literature develops these mechanisms in relative isolation, emphasizing portfolio heterogeneity, rate sensitivity, or governance discipline separately,

rather than within an integrated analytical framework. Asset selection, funding structure, and macro-financial conditions may interact through dynamic feedback channels, thereby shaping the evolution of BDC risk and return over time. Identifying and explaining these interaction mechanisms, and their implications for persistent cross-sectional heterogeneity and cyclical revaluation in the industry remains an important issue for further analysis.

2 Hypothesis Development

The core assets of BDCs consist mainly of loans to SMEs and private companies, or other forms of private debt investments. Unlike publicly traded bonds or standardized loans, these assets are typically less liquid, exhibit less transparency, and make it more difficult for external investors to directly observe the true operating conditions and risk characteristics of the borrowers. Therefore, although the NAV of BDCs can to some extent reflect the performance of their underlying assets, whether market prices can consistently and efficiently track changes in NAV still requires further empirical examination.

Existing research suggests that the private debt market is inherently characterized by substantial heterogeneity and specialization. [Carey et al., 1998] find that finance companies tend to serve borrowers with higher leverage and greater risk, indicating that the identification and pricing of such assets rely largely on professional judgment rather than publicly available information. For BDCs, which primarily invest in private debt, this implies that their underlying assets are inherently opaque. As a result, market investors may not be able to understand the information embedded in NAV as accurately and timely as they would when evaluating publicly traded securities.

Based on this, changes in interest rates may further weaken the extent to which market prices reflect the fundamentals captured by NAV. When interest rates rise, the discount rate used in asset valuation increases accordingly, leading to a decline in the present value of future cash flows. At the same time, rising interest rates are often associated with greater uncertainty and higher required risk compensation, causing asset pricing to rely more heavily on discount rates and risk premia, while relatively diminishing the role of underlying cash flows.

Prior research also suggests that in environments characterized by heightened financial frictions or increased risk management pressures, financing conditions and risk premia play a more prominent role in asset pricing. [Chen et al., 2017] find that, following the financial crisis, large banks contracted their lending to small businesses, leading to persistently higher financing costs in affected regions. Similarly, [Acharya et al., 2018] show that under stress testing constraints, banks increase loan spreads and reduce credit supply to riskier borrowers. These findings indicate that when credit conditions tighten, market pricing tends to rely more heavily on risk compensation and financing costs, rather than being driven solely by underlying cash flows. This mechanism is particularly relevant for BDCs that primarily invest in private debt. Given the opacity of their underlying assets, investors face greater difficulty in directly assessing

future cash flows, and therefore rely more on discount rates and risk premia for valuation. As a result, during periods of rising interest rates, asset pricing may be increasingly influenced by changes in discount rates rather than the performance of the underlying assets themselves, thereby weakening the extent to which market prices reflect NAV.

Based on the analysis, we propose the following hypothesis:

H1: As interest rates increase, the sensitivity of BDC market returns to NAV returns decreases.

Furthermore, this effect may not be uniform across all BDCs. Corporate governance quality may play an important moderating role. For BDCs holding illiquid and non-public assets, investors' willingness to rely on NAV depends not only on the underlying assets, but also on whether the governance structure enhances the credibility of information disclosure and the valuation process. In the context of direct lending and BDCs, governance mechanisms and market discipline play a critical role in risk selection and the information environment [Davydiuk et al., 2024c]. This suggests that even when asset characteristics are similar, differences in governance structures may lead to systematic variation in how investors perceive information quality and risk. Therefore, BDCs with higher governance quality are more likely to establish effective monitoring mechanisms and a more credible valuation environment. As a result, even during periods of rising interest rates and heightened uncertainty, they are more likely to maintain investor trust in NAV. For such firms, the deviation between market prices and NAV is likely to be smaller, and the weakening effect of rising interest rates on the NAV market return relationship is expected to be less pronounced.

H2: For BDCs with higher corporate governance quality, the weakening effect of rising interest rates on the relationship between NAV and market returns is less pronounced.

In addition, differences in asset characteristics across BDCs may lead to significant variation in this effect. For BDCs with higher asset opacity, their underlying investments typically rely on non-public information and subjective valuation judgments, making it difficult for external investors to directly observe their true risk profiles. [Carey et al., 1998] point out that lending decisions in the private debt market exhibit strong specialization, and asset valuation depends heavily on intermediaries' ability to acquire and interpret information. When interest rates rise and market uncertainty increases, investors may find it even more difficult to accurately assess the intrinsic value of such assets, thereby reducing their reliance on NAV and instead adopting more conservative pricing approaches. Consequently, compared to BDCs with more transparent assets, those with higher asset opacity are more likely to experience deviations between market prices and fundamentals.

H3: For BDCs with higher asset opacity, the weakening effect of rising interest rates on the relationship between NAV and market returns is stronger.

3 Research Design

$$R_{i,t}^{MKT} = \alpha + \beta_1 R_{i,t}^{NAV} + \beta_2 \Delta Rate_t + \beta_3 (R_{i,t}^{NAV} \times \Delta Rate_t) + \gamma X_{i,t} + \epsilon_{i,t} \quad (1)$$

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